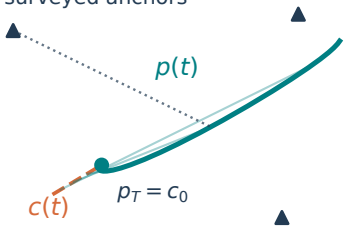
(a) Window model

$p(t) = c(t) + R_z(\psi)d(t)$

surveyed anchors



Known IMU shape $d(t)$; unknown $c(t), \psi$

(b) Candidate and refinement

Squared-range lift
$y = Bv + Dc + Cu$

Nuisance projection
$\bar{y} = P_L W y, \quad \bar{C} = P_L W C$

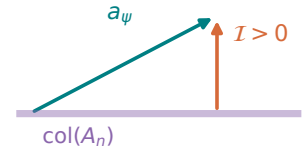
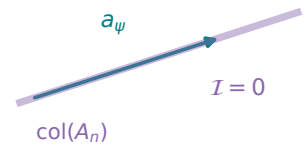
Circle candidate
$u = [\cos\psi, \sin\psi]^\top$

CV fit from two seeds
↓
retain best translation

Raw-range Huber refinement
endpoint p_T, v_T, ψ for the InEKF

(c) Heading information

Yaw component survives



Yaw lies in nuisance span